

Print at 100% / Actual size, SINGLE-sided. Crease along the grey line, fold so both printed faces are outward, glue, then cut. This bar must measure exactly 80 mm.

[illegible]

FIRST SOLVE

Now here? Do the walkthrough once at cubepath-six.vercel.app/learn. This card is the memory jog.

NOTATION

R U F L D B = turn that face clockwise, looking at from outside the cube.
 = counter-clockwise = 2 = half turn. Lowercase r f = that face plus the layer behind it, turning together.
 y = spin the whole cube left, white staying down. A whole cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm you hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out.

Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until its above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT

RUR'



white sticker FRONT

L'ULU



white sticker UP

turn the first one once to knock it out, then look again

RUR'

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes in.



goes RIGHT

R'UR' y L'ULU



goes LEFT

U' L'ULU y' RUR'U'

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

■ RUR' sexy move
 ■ RUR'U Sume
 ■ R'FRP' sledgehammer
 gap = change grip

[illegible]

ONE-LOOK PLT

ADJACENT CORNER SWAP

exactly one face shows headlights



• **T** L headlights: pairs B-left F-left
RUR'U' R'F R2UR'U' RUR'F'

J L solid: pairs B-right F-left front
x R2RF' RU2 rUr U2

J b L solid: pairs B-left F-right back
RUR'F' RUR'U' R'F R2UR'

A L headlights: pairs F-right B-front
x L2D2 L'U'L D2 L'U'L'

A b L headlights: pairs B-right R-back
x L2D2 LUL' D2 LU'L

F L solid
R'U'F' RUR'U' R'F R2UR'U'

ADJACENT CORNER SWAP

continued



R L headlights: pairs F-left
RU'U' R'UR DR'URD' R'U2R'

R b L headlights: pairs B-left
R2F RURU'R' F'R U2R'U2R

G L headlights: pairs F-right
R2UR'UR'U' RU'R2 U'D R'URD'

G b L headlights: pairs R-back
RU'R U'D' R2UR'UR' RU'R2D

G e L headlights: pairs B-right
R2UR'UR'U' RU'R2 U'D' RU'RD

G d L headlights: pairs R-front
RU'R U'D' R2UR'UR' UR'UR2D'

One face shows headlights, two corners of that face match. Hold a drive, run, then turn the top to finish.

RUR'U' means: **RUR'U'** same, **R'F'F'** *skolechammer*, **U** = *chamber* *into*

The diagram illustrates the Vigenère cipher encryption process. It shows how a plaintext message is combined with a key stream to produce a ciphertext.

Key Stream: RUFRR RUFRR RUFRR

Plaintext: START HERE

Ciphertext: RUFRR RUFRR RUFRR

Encryption Process:

The encryption process involves combining the plaintext and the key stream using the Vigenère square. The key stream is derived from the key 'RUFRR' and the plaintext 'START HERE'.

Step 1: Plaintext and Key Stream

Plaintext: S T A R T H E R E
Key Stream: R U F R R R U F R R

Step 2: Encryption

The encryption process involves combining the plaintext and the key stream using the Vigenère square. The key stream is derived from the key 'RUFRR' and the plaintext 'START HERE'.

Step 3: Ciphertext

Ciphertext: R U F R R R U F R R

Decryption Process:

The decryption process involves reversing the encryption process. The ciphertext is combined with the key stream to recover the plaintext.

Step 1: Ciphertext and Key Stream

Ciphertext: R U F R R R U F R R
Key Stream: R U F R R R U F R R

Step 2: Decryption

The decryption process involves reversing the encryption process. The ciphertext is combined with the key stream to recover the plaintext.

Step 3: Plaintext

Plaintext: S T A R T H E R E

TWO-LOOK OLL CROSS yellow edges	OLL CORNERS continued
 Line has left-to-right FRUR'U'F'	 Headlights $\approx$ 2 yellow; headlights at the back 2x2RU'U RD'R'U'2R'
 Hook hook pointing front-right fRUR'U'f'	 2x Headlights $\approx$ 0 yellow; headlights both sides RU'R'U' RU'R' URU'2R'
OLL CORNERS yellow face	Chameleon $\approx$ 2 yellow next to each other
 Sunes $\approx$ 1 yellow; rest clockwise RU'R'U' RU'2R'	 Bowtie $\approx$ 2 yellow diagonal F'rURU'r' FR
 Anti-Sunes $\approx$ 1 yellow; rest counter-clockwise RU'2R' U'RUR'	FALLBACK The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked, there is the worst. Top not all yellow and nothing matches? Run Sunes and read again.
 Pi $\approx$ 0 yellow; headlights left fRUR'U'f' FRUR'U'F'	

[illegible]

BIG CUBES – BIG CURES AND THE MAP

ANXEXES 4x4 and 5x5
Reduce: build the 12 centers → solve it as a 3x3. Rw = 2 layers wide; 3Rw = 3 layers w/ 2R = 2nd LAYER ONLY – never widen it.

last 2 edges
4x4 PLL parity $R'w^2 \quad U' \quad R' \quad U \quad R' \quad F' \quad R' \quad F'$ slice out, run it, slice back – both cubes
 $2R^2 \quad U^2 \quad R^2 \quad Uw^2 \quad 2R^2 \quad Uw^2$ 2 edge pairs swapped - half the time

4x4 OLL parity 1-edge pair flipped - half the time
 $Rw \quad U^2 \quad x \quad Rw \quad U^2 \quad Rw \quad U^2 \quad Rw' \quad U^2 \quad Lw \quad U^2 \quad Rw' \quad U^2 \quad Rw \quad U^2 \quad Rw' \quad U^2 \quad Rw'$
5x5 edge parity 1-edge pair flipped - half the time *one move differs*; 5x5 has no PLL parity
 $Rw \quad U^2 \quad x \quad Rw \quad U^2 \quad Rw \quad U^2 \quad 3Rw^2 \quad U^2 \quad Lw \quad U^2 \quad Rw' \quad U^2 \quad Rw \quad U^2 \quad Rw' \quad U^2 \quad Rw'$
1st last layer, before OLL and PLL – both parity algorithms move center

NOTATION

 Each letter stands for that face clockwise,
from outside the cube – so from your point of view.
Your seat L, D and B look counter-clockwise –> counter-clockwise, 2 = half turn.
M = middle slice, turning the way L turns, 3x, y,z = the whole cube, turning the way R, U and F turn. Lowercase f if that face plus the layer behind it, turning together.

WORDS
algorithm = a fixed sequence of turns - trivial - a short algorithm your hands run as one unit - sexy move = the R U R' U' trigger -
the R' F R F' trigger, parity = a case 3x3 cannot have an edge pair = two edges glued into one, on a 4x4 or 5x5

$\text{R} \rightarrow \text{RU} \text{RL}$ sexy moves $\text{R}' \rightarrow \text{RU} \text{L}'$ $\text{F} \rightarrow \text{FR} \text{FL}$ slide moves no c-change